

ENGR 40 Truss Design and Analysis Document

Design Screenshots:

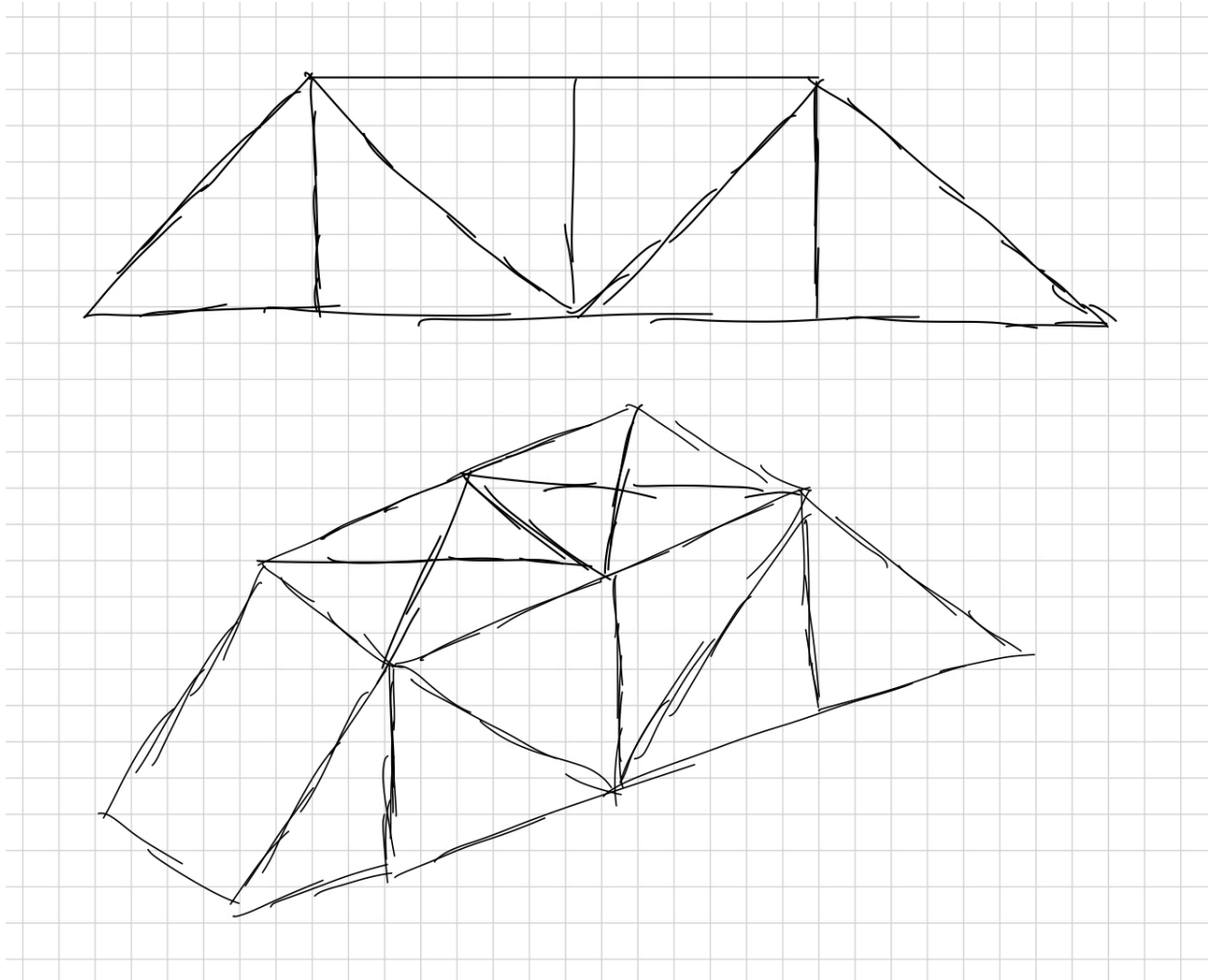


Figure 1: Initial Sketched Design

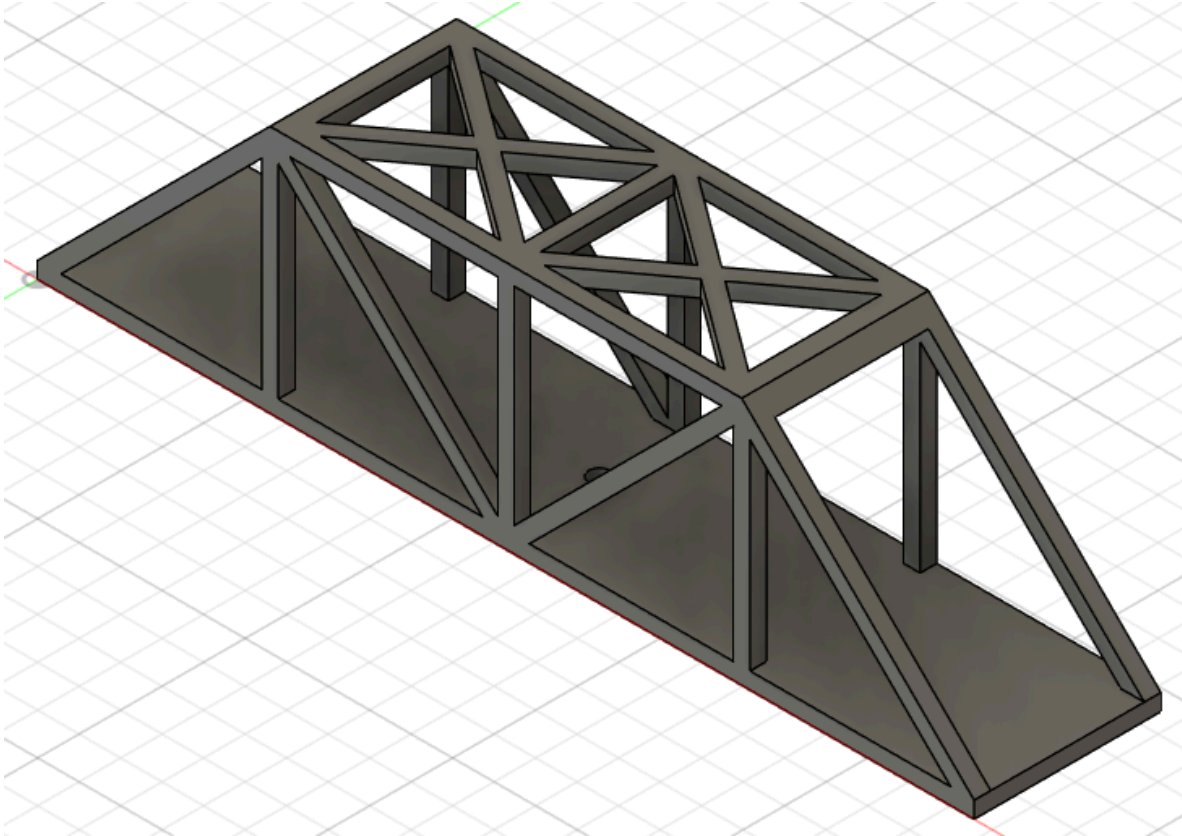


Figure 2: Isometric View of Truss Design.

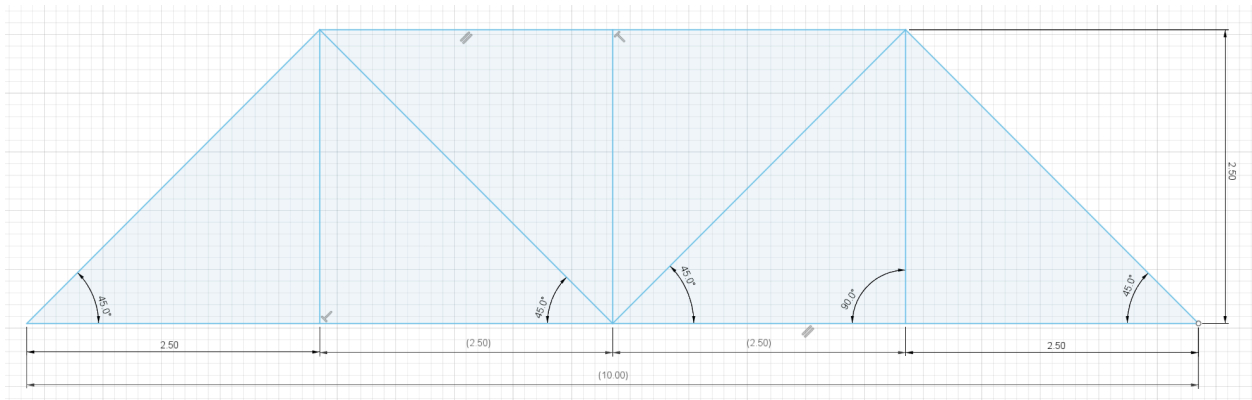


Figure 3: Dimensions of Truss Design.

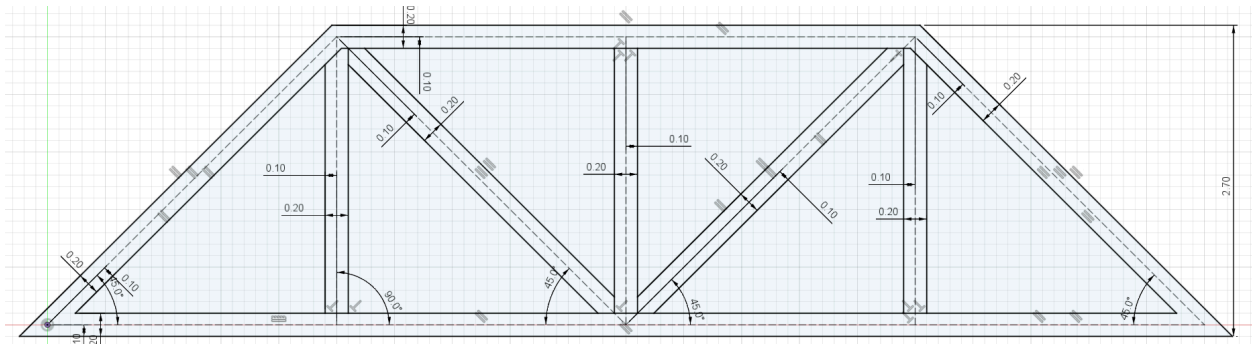
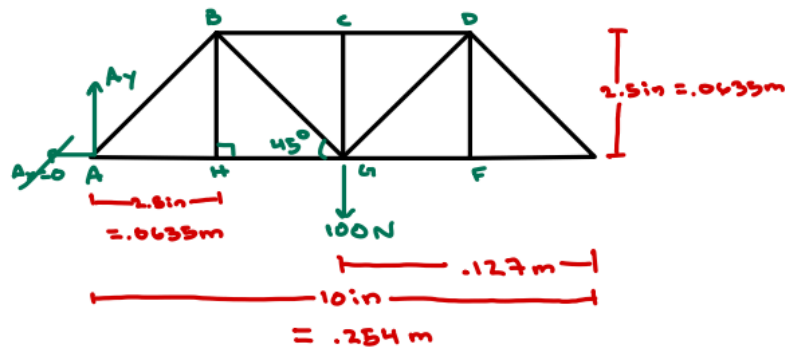


Figure 4: Actual Dimensions of Truss Design.

Methods of Joints Analysis:

Method of Joints Analysis



EoC E: $\sum M_E = 0: 100\text{N}(.127\text{m}) - A_y(.254\text{m}) = 0$
 $A_y = 50\text{N}$

A:



$$\sum F_y = 0: 50\text{N} + F_{AB} \sin(45^\circ) = 0$$

$$F_{AB} = -70.7\text{N} = 70.7\text{N}(C)$$

$$\sum F_x = 0: F_{AB} \cos(45^\circ) + F_{AH} = 0$$

$$F_{AH} = -(-70.7\text{N}) \cos(45^\circ)$$

$$F_{AH} = 50\text{N}(T)$$

H:



$$\sum F_y = 0: F_{HB} = 0$$

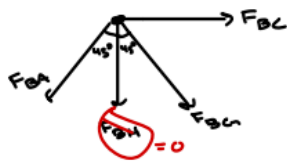
$$\sum F_x = 0: F_{HG} - F_{HA} = 0$$

$$F_{HG} = F_{HA}$$

$$F_{HG} = 50\text{N}(T)$$

Figure 5: Methods of Joints Analysis.

B:



$$\sum F_y = 0: -F_{BA} \cos(45^\circ) - F_{BH} \cos(45^\circ) = 0$$

$$F_{BH} = -F_{BA}$$

$$F_{BH} = -(-70.7 \text{ N})$$

$$F_{BH} = 70.7 \text{ N (T)}$$

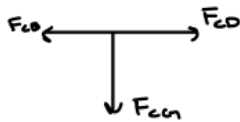
$$\sum F_x = 0: F_{BC} + F_{BG} \cos(45^\circ) - F_{BA} \cos(45^\circ) = 0$$

$$F_{BC} = F_{BA} \cos(45^\circ) - F_{BG} \cos(45^\circ)$$

$$F_{BC} = -70.7 \text{ N} \cos(45^\circ) - 70.7 \text{ N} \cos(45^\circ)$$

$$F_{BC} = 100 \text{ N (C)}$$

C:



$$\sum F_x = 0: F_{CD} - F_{CB} = 0$$

$$F_{CD} = F_{CB}$$

$$F_{CD} = 100 \text{ N (C)}$$

$$\sum F_y = 0: F_{CB} = 0$$

G:



$$\sum F_y = 0: F_{GB} \sin(45^\circ) - 100 + F_{GD} \sin(45^\circ)$$

$$F_{GD} = \frac{100 - 70.7 \sin(45^\circ)}{\sin(45^\circ)}$$

$$F_{GD} = 70.7 \text{ N (T)}$$

$$F_{GB} = F_{GD}$$

$$\sum F_x = 0: F_{GF} + F_{GD} \cos(45^\circ) - F_{GH} - F_{GB} \cos(45^\circ) = 0$$

$$F_{GF} = F_{GH}$$

$$F_{GF} = 50 \text{ N (T)}$$

F:



$$\sum F_y = 0: F_{FD} = 0$$

$$\sum F_x = 0: F_{FE} - F_{FH} = 0$$

$$F_{FE} = 50 \text{ N (T)}$$

E:



$$\sum F_x = 0: -F_{ED} \cos(45^\circ) - F_{EF} = 0$$

$$F_{ED} = \frac{-F_{EF}}{\cos(45^\circ)} = -70.7 \text{ N}$$

$$F_{ED} = 70.7 \text{ N (C)}$$

Figure 6: Methods of Joints Analysis Continued.

Percent Loads

$$\% \text{ Load} = \frac{\text{Force of member}}{\text{Total Force of members}} \times 100\%$$

<u>Individual Forces</u>		<u>% Load</u>
$F_{AB} = 70.7 \text{ N (C)}$	→	$F_{AB} = 10.4\%$
$F_{AH} = 50 \text{ N (T)}$	→	$F_{AH} = 7.3\%$
$F_{HB} = 0$	→	$F_{HB} = -$
$F_{HG} = 50 \text{ N (T)}$	→	$F_{HG} = 7.3\%$
$F_{BG} = 70.7 \text{ N (T)}$	→	$F_{BG} = 10.4\%$
$F_{BC} = 100 \text{ N (C)}$	→	$F_{BC} = 14.6\%$
$F_{CD} = 100 \text{ N (C)}$	→	$F_{CD} = 14.6\%$
$F_{CH} = 0$	→	$F_{CH} = -$
$F_{HD} = 70.7 \text{ N (T)}$	→	$F_{HD} = 10.4\%$
$F_{HE} = 50 \text{ N (T)}$	→	$F_{HE} = 7.3\%$
$F_{ED} = 0$	→	$F_{ED} = -$
$F_{FE} = 50 \text{ N (T)}$	→	$F_{FE} = 7.3\%$
$F_{FD} = 70.7 \text{ N (C)}$	→	$F_{FD} = 10.4\%$

} F_{BC} & F_{CD} are critical members

Total Force = 682.8 N

Figure 6: Methods of Joints Analysis Continued.

Group Contributions:

Tianwei Xi:

- Completed CAD Design of the Bridge using Fusion 360.
- Generated .gcode File to 3D Print.
- Worked on the Images Portion of the Deliverable Document.

Aaron Acevedo:

- Initial Truss Design/Brainstorming.
- Worked on the Images and Method of Joints Analysis portion of the Deliverable Document.
- Printed the Bridge Design Using the 3D Printer in Class.

Lance Aquino:

- Method of Joints Analysis.
- Initial Truss Design/Brainstorming.
- Worked on the Method of Joints Analysis portion of the Deliverable Document.
- Printed the Bridge Design Using the 3D Printer in Class.

Summer Schwartz:

- Method of Joints Analysis.
- Worked on the Method of Joints Analysis portion of the Deliverable Document.
- Printed the Bridge Design Using the 3D Printer in Class.